

CONTACT INFORMATION	1 (217) 305-2300 noejnava2@gmail.com https://noejn2.github.io/ Citizenship: United States of America and México	
POSITIONS	Seize — Institute of Consulting and Business Solutions 2025 - <i>present</i> Economic Modeling Advisor Riyadh, Saudi Arabia Develop and deploy computable general equilibrium (CGE) and gravity models to assess trade agreements and policy reforms. Build institutional modeling capacity, including AI-powered natural-language interfaces that let stakeholders query model results and run scenarios without writing code. Pair data-driven and economics-driven analysis to inform policy decisions, and deliver capacity-building workshops for government staff. Independent Consulting 2025 Economic Modeling Consultant Chicago, IL USDA Economic Research Service 2022 - 2025 Research Agricultural Economist Market and Trade Economics Division, Agricultural and Policy Models branch	
EDUCATION	Ph.D., Agricultural and Applied Economics 2021 University of Illinois Urbana-Champaign, IL Advisor: William Ridley Dissertation: Three Essays on Agricultural and Consumer Economics, <i>Outstanding Doctoral Dissertation Award</i> Committee: William Ridley (chair), Sandy Dall’Erba, Yilan Xu, and Rafael Garduño-Rivera B.A., Mathematical Economics 2016 California State University, San Bernardino, San Bernardino, CA Advisor: Kazim Konyar <i>Summa Cum Laude</i>	
RESEARCH INTERESTS	Methods Bayesian data analysis General equilibrium models Structural econometrics Machine learning & AI for policy	Fields Agricultural production Consumption economics Food systems
JOURNAL ARTICLES	1. <i>Land competition and poverty effects from the Mexico’s proposal to ban Genetically Engineered corn.</i> 2024 American Journal of Agricultural	

- Economics** 106(3): 1300-1325.
with Jayson Beckman, Angelica Williams, and Steven Zahniser.
doi:10.1111/ajae.12463
2. *Losing Ground: Measuring the Welfare Effects of Retail Food Price Inflation during the COVID-19 Pandemic on Mexican Household.* 2024 **Journal of Agriculture and Food Industrial Organization** 22 (2): 143-154.
with Benjamin D.K. Wood, and Rafael Garduño-Rivera.
doi:10.1515/jafio-2023-0041
 3. *Estimation of the Value-Added/Intermediate Input Substitution Elasticities Consistent with the GTAP Data.* 2023. **Journal of Global Economic Analysis** 8(2): 134-158.
with Jayson Beckman, and Maros Ivanic.
doi:10.21642/JGEA.080204AF
 4. *A model of the U.S. food system: What are the determinants of the state vulnerabilities to production shocks and supply chain disruptions?* 2023. **Agricultural Economics** 54(1): 95-109. ¹ ²
with William Ridley, and Sandy Dall’Erba.
doi:10.1111/agec.12750
 5. *The impact of taxing sugary-sweetened beverages in México: A censored QUAID demand system approach.* 2022. **Journal of Agricultural and Applied Economics Association** 1(1): 18-32.
with Diansheng Dong.
doi:10.1002/jaa2.6
 6. *U.S. interstate trade will mitigate the negative impact of climate change on crop profit.* 2021. **American Journal of Agricultural Economics** 103(5): 1720-1741.
with Sandy Dall’Erba, and Zhangliang Chen.
doi:10.1111/ajae.12204
 7. *The impacts of agroforestry on agricultural productivity, ecosystem services, and human well-being in low- and middle-income countries: An evidence and gap map.* 2020. **Campbell Systematic Reviews** 16(1): e1066.
with Daniel C. Miller, Pablo J. Ordoñez, Sarah E. Brown, Samantha Forrest, Karl Hughes, and Kathy Baylis.
doi:10.1002/cl2.1066
 8. *The conduits and barriers to reentry for formerly incarcerated individuals in San Bernardino.* 2018. **Journal of Prison Education and Reentry** 5(1): 2-17.
with Annika Anderson, and Patricia Cortez.
doi:10.25771/sdf0-1631

¹Most Outstanding Second Year Research Paper at University of Illinois at Urbana-Champaign-ACE 2020.

²The Charles M. Tiebout Prize in Regional Science 2021.

- BOOK CHAPTERS
1. *Climate and Weather Impacts on Agricultural Productivity*. 2025. **Encyclopedia of Energy, Natural Resource, and Environmental Economics**, 2e. with Eric Njuki, and Boris E Bravo-Ureta.
- U.S.
DEPARTMENT OF
AGRICULTURE,
ECONOMICS
RESEARCH
SERVICE
1. *International Income and Price Elasticity Estimates: An Update* 2025 **Technical Bulletin** TB-1971. with Jacob Abrehe Zereyesus, Tian Xia, Xianghong Li, and Lila Cardell
 2. *Potential Budgetary Impacts of Climate Change on the Pasture, Rangeland, and Forage Insurance Plan* 2024 **Economic Brief** EB-38: 25pp. with Dylan Turner, Katherine Baldwin, Jayson Beckman, Francis Tsiboe, and Kate Vaiknoras.
 3. *ERS Research Models Future Effects of Climate Change on Corn and Soybean Yields, Production, and Exports* 2024 **Amber Waves** 1545-8741. with Jayson Beckman. ERS Amber Waves, March 04, 2024
 4. *Estimating the market and trade implications from climate-Induced corn and soybean yield changes for the U.S.* 2023 **Economics Research Report** ERR-324: 31pp. with Jayson Beckman, and Maros Ivanic.
 5. *Warming temperatures in U.S. Corn Belt expected to continue into next decade.* Charts of Note, December 5, 2023
- WORKING
PAPERS
1. *Tracing U.S. county-to-county domestic trade flows in grains.* with William Ridley, Sandy Dall’Erba, Jian Zou, Youning Chen, and Christa Court. AAEA discussion paper
 2. *The Impact of International Demand for Cereals on U.S. Crop Land Appreciation: A Model Implied Instrumental Variable Approach* with William Ridley, and Stephen Morgan. AAEA discussion paper
 3. *Does trade openness affect global grain supply response?* with Jacob Zereyesus, Jennifer Kee, Stephen Morgan, Nathan Hendricks, Michael Johnson, Lila Cardell, Jeremy Jelliffe, and Walter Acpangan. AAEA discussion paper
 4. *Estimation of the factor transformation elasticities consistent with the GTAP data.* with Jayscon Beckman, and Maros Ivanic.
 5. *Incorporating the Land Allocation System into the GTAP framework* with Kristiina Ala-Kokko, Jayson Beckman, and Maros Ivanic. GTAP discussion paper

6. *Heterogenous patterns of crop yield growth stagnation across U.S. counties in the next decade*
Solo paper.
7. *Agro-climatic patterns for the Corn Belt towards the middle of the century.*
with Eric Njuki, Jayson Beckman, and Maros Ivanic.
8. *Revisiting the impact of climate change on agriculture through spatially-varying and place-tailored Ricardian estimates.*
with Sandy Dall’Erba, Chang Cai, and A. Stewart Fotheringham.
CREATE discussion paper

PAPERS IN
PREPARATION

1. *Is agricultural insurance a safety net or a net of complacency amidst adaptation to extreme weather events?*
with Francis Tsiboe, Dylan Turner, Katherine Baldwin, and Sharon Raszap.
2. *Designing policies to achieve healthy and sustainable diets*
with Grace Melo-Guerrero.

SOFTWARE
& PUBLIC TOOLS**R & Julia packages**

gravityGE — one-sector Armington-CES gravity model with general equilibrium effects (CRAN; Julia port *GravityGE.jl* on JuliaHub).
censoredAIDS — maximum-likelihood estimation of censored (Quadratic) Almost Ideal Demand Systems (CRAN).

Interactive dashboards

U.S. Domestic Agricultural Trade — interstate agricultural trade flows with gravity / general-equilibrium policy simulation.
 Fertilizer Flows — visualization of global fertilizer trade flows.

AI tools

AI Paper Referee — large-language-model auditor that reviews applied-economics writing against published standards.

FELLOWSHIPS,
GRANTS AND
SCHOLARSHIPS

- The impact of climate change on agricultural production, productivity, and trade (\$403,152) 2023
- Improving the capabilities of global trade models to analyze changes in land use/land cover and green house emissions (\$280,000) 2023
- A global agricultural supply-demand-climate-trade model calibrated to Baseline (\$375,000) 2022
- Towards a data hub of global daily weather forecasts for the ERS (**PI**) (\$30,000) 2022
- ERS’ international supply forecasting (\$30,000) 2022
- International food consumption patterns-implications for global food security and trade (\$25,000) 2022
- The Charles M. Tiebout Prize in Regional Science (\$1,000) 2021
- Cleo Fitzsimmons Award (\$500) 2020
- Agricultural and Consumer Economics Department Grant (\$400) 2019

	• Marshall Scott Fellowship (\$4,877.85) 2018 • Brockson Fellowship (\$3,902.28) 2017 • The Proactive Recruitment in Introductory Science and Mathematics Travel Grant (\$1,000) 2016 • The Proactive Recruitment in Introductory Science and Mathematics Research Grant (\$2,000) 2015 • The John & June Kennedy Economics Scholarship (\$1,500) 2015
AWARDS AND HONORS	• Outstanding Doctoral Dissertation Award, University of Illinois at Urbana-Champaign-ACE 2021 • The Charles M. Tiebout Prize in Regional Science 2021 • Farm foundation Agricultural Scholar 2021 • Most Outstanding Second Year Research Paper, University of Illinois at Urbana-Champaign-ACE 2020
TEACHING AND MENTORSHIP EXPERIENCE	Mentor at ERS Summer internship Kristiina Ala-Kokko (Kansas State University) 2022 Teaching Assistant ACE 240: Personal Financial Planning FA19 Instructor: Paul B Stoddard, Senior Lecturer <i>Lecture:</i> 160 students University of Illinois at Urbana-Champaign, IL Undergraduate Credit: 4 Hours. ACE 231: Food and Agribusiness Management SP18 Instructor: Paul B Stoddard, Senior Lecturer <i>Lecture:</i> 160 students University of Illinois at Urbana-Champaign, IL Undergraduate Credit: 4 Hours. ACE 341AMM: Issues in Agribusiness Markets and Management FA17 Instructor: Paul B Stoddard, Senior Lecturer <i>Lecture:</i> 20 students University of Illinois at Urbana-Champaign, IL Undergraduate Credit: 3 Hours.
RESEARCH IN THE NEWS	1. <i>Climate Impacts on Corn and Soybean Yields by Mid-2030s</i> in Progressive Farmer 2. <i>“Climate change is real;” Researchers warn extreme weather events could impact Illinois agriculture</i> in WCIA.com 3. <i>Trade Insurance</i> in Jayson Lusk, Food and Agricultural Economist, personal blog.

RESEARCH ASSISTANTSHIPS	Interstate Trade as Mitigating Factor for the Effects of Climate Change on Illinois Agriculture January 2020 - December 2021 How do the interstate trade flows of Illinois farm products respond to extreme weather events? University of Illinois at Urbana-Champaign, IL Supervisor: Prof. William Clifton Ridley
WORK EXPERIENCE	Intern Department of Agriculture, U.S., Economic Research Service Summer 2021 Market & Trade Economics Division Agricultural Policy and Models Branch Kansas City, MO Intern Department of Agriculture, U.S., National Agricultural Statistics Service 2016 - 2017 Environmental, Economics and Demographics Washington, DC Intern Department of Agriculture, U.S., Rural Development 2015 - 2016 Rural Housing Service Washington, DC
SELECTED PRESENTATIONS	The Impact of International Demand for Agricultural Products on U.S. Cropland Appreciation: A Model-Implied Instrumental Variable Approach 2024 Agricultural & Applied Economics Association (AAEA) The Impact of International Demand for Agricultural Products on U.S. Cropland Appreciation: A Model-Implied Instrumental Variable Approach 2023 The International Agricultural Trade Research Consortium (IATRC) Heterogeneous Patterns of Crop Yield Growth Stagnation across U.S. Stagnation across U.S. Counties in the Next Decade 2023 Agricultural & Applied Economics Association (AAEA) Track Session: Studying Climate Change Impacts from Different Methodological Perspectives The probabilistic role of climate change on crop yield potential 2022 FCSM Research & Policy Conference (FCSM) The impact of taxing sugar-sweetened beverages in México: A censored QUAID demand system 2022 Agricultural & Applied Economics Association (AAEA) Track Session: Identification in Demand Estimations A model of U.S. domestic crop trade to study state's food vulnerability to supply chain disruptions ³

³Previously titled *Who benefits from local agriculture?*

Agricultural & Applied Economics Association (AAEA)	2021
Western Regional Science Association (WRSA)	2021
North American Regional Science Council (NARSC)	2020

**Revisiting the impact of climate change on
agriculture through spatially-varying and place-tailored
Ricardian estimates**

North American Regional Science Council (NARSC)	2021
Workshop RED in Mexico: Challenges of a New Era	2018

REFERENCES

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Professor, Department of Agricultural and Consumer Economics
University of Illinois Urbana-Champaign, IL
wridley@illinois.edu

Sandy Dall’Erba, Ph.D.
Professor, Department of Agricultural and Consumer Economics
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Rafael Garduño-Rivera, Ph.D.
Professor, School of Economic and Administrative Sciences
Universidad Panamericana, México
rgardunor@up.edu.mx

PROGRAMMING
& DEPLOYMENT

Julia, L^AT_EX, Markdown, Python, R, Stan, Stata, & SQL.
Docker, AWS, Azure, Shiny Server, Git & GitHub Actions (CI/CD).
LLM/AI tooling for natural-language model interfaces and research workflows.

LANGUAGES

English & Spanish.

SERVICE AND
PROFESSIONAL
ACTIVITIES

Journal Refereeing

American Journal of Agricultural Economics; Economic Modelling; Discover
Food;s Environmental Research Letters; Journal of Agricultural and Applied
Economics Association; Nature Food; Organisation for Economic Co-Operation
and Development (OECD); U.S. Global Change Research Program (USGCRP);
World Development.

USDA - Economics Research Service

Member, ERS Modeling Team Committee
Member, ERS Brownbag Series Committee

Abstract Reviewer

AAEA Annual Meeting, 2025.
AAEA Annual Meeting, 2024.